

Learning Journal

Reference Script:

Jigsaw Puzzle Game:

<https://youtu.be/9mSwbMiV3IU?si=Qg5XJMsUnMqd9Pke>

Work Log:

[Work Log.xlsx](#)

Week 2:

November 11, 2025

During, the first week of starting the project I set up a GitHub account and made my first repository. Then, I created a folder and files to make individual sections of tutorials. Afterwards, I made created a Unity file with the version to produce a Jigsaw Puzzle Game. Also, I created an Excel sheet to log in my work whenever I work on my Unity project.

I thought about what dimension I wanted my prototype to be in editor version of 6000.2.8f1. Finally, I decided to make it 2D as I was thinking in the future I would like to make my game about puzzles. I created a note to see what kind of mechanics I can implement into my tutorials such as a Drag and Drop, Generate Maze and Path Finder. Then, I looked through YouTube videos to see if I could take some of the ideas.

Programming Module
2D

Memory game
Jigsaw game
Maze

Main Mechanics to implement:
Timer
Generate Maze
Path Finder
Drag and Drop

November 12, 2025

Today, I started my Unity project and called the script called Room to make a maze. Then, I made a 2d square as I called each side BOTTOM, RIGHT, UP and LEFT and group them into an empty and call it 'Room'. The script will manage the walls inside the maze and how it'll interact when the maze is generated.

Afterwards, I created a script named GenerateMaze and created a variable which removes the walls as it generates a randomised a maze.

November 15, 2025

Today, I learnt about the ways to implement the Path Finder as I discovered about three different algorithms. The first algorithm I learnt about is Dijkstra's Algorithm which explores the path uniformly, considering all neighbouring nodes without heuristic guidance towards the goal. The second algorithm I learnt about is A* Algorithm uses a heuristic function to guide the search towards the goal as it combines the cost to reach a node from the start and estimate cost to reach the goal from that node. A good sign of a good heuristic is the algorithm will significantly reduce the search space and better performance. Lastly, I learnt about the Greedy Best- First Search as it's a uniformed search algorithm that explores a graph by prioritising node based on their heuristic. However, in contrast to the Dijkstra, this algorithm it's faster to find a solution id the heuristic function is well-designed but it's not a guaranteed optimality.

Afterwards, I started my Unity project and called the script Path Finder as it'll find the best path between point A and B. This will be more interactive when pairing it with the generate maze.

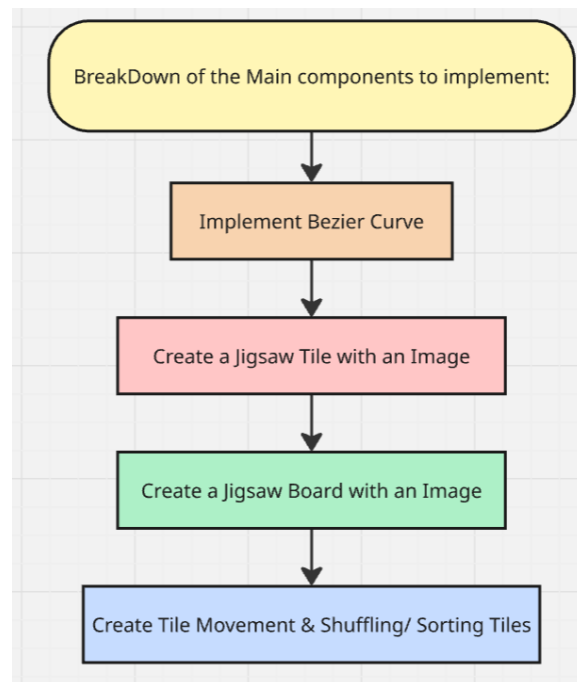
November 16, 2025

I continued creating a Path Finder as it was almost completed. A thing I struggled in with the script was clicking from A to B and have a path finder flow as it would have just not worked and had the two points but no movement.

Week 3

November 18, 2025

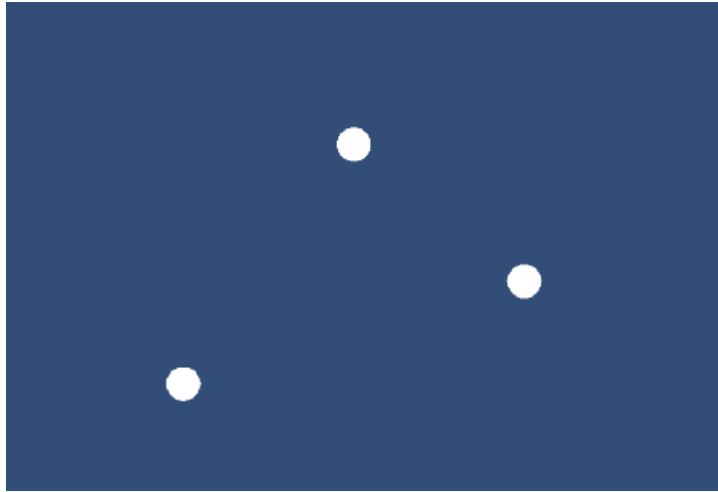
Ultimately, I decided to scrap the Generate Maze and Path Finder. I went on Miro to create a workflow and stated which 4 main components I wanted to focus on. I based it off on the ideas I made during the first week of things I wanted to make.



November 19, 2025

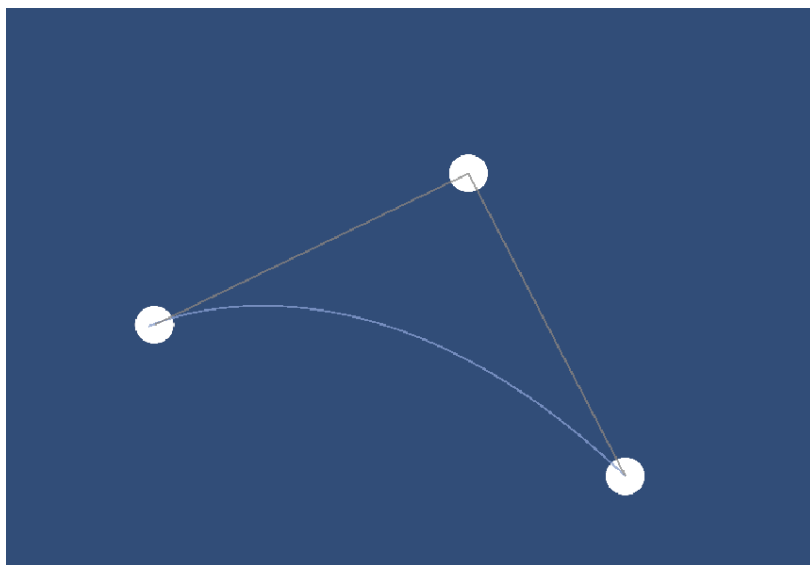
Today, I learnt about the implementation of Bezier curve and what a Bezier curve is. Also, I learnt about the equation.

Afterwards, I started up my Unity project and created a script called Bezier and another one called Bezier Curve. For the Point script I applied a prefab of a 2d circle and 2D collider and attached the Point script onto it. Then, I continued with the Bezier script as I used the Bezier Curve script as reference. When you press play it'll show three points that will later relate to a line.



November 21, 2025

Today, I continued with the Bezier and updated so the Bezier curve lines up with the three points.

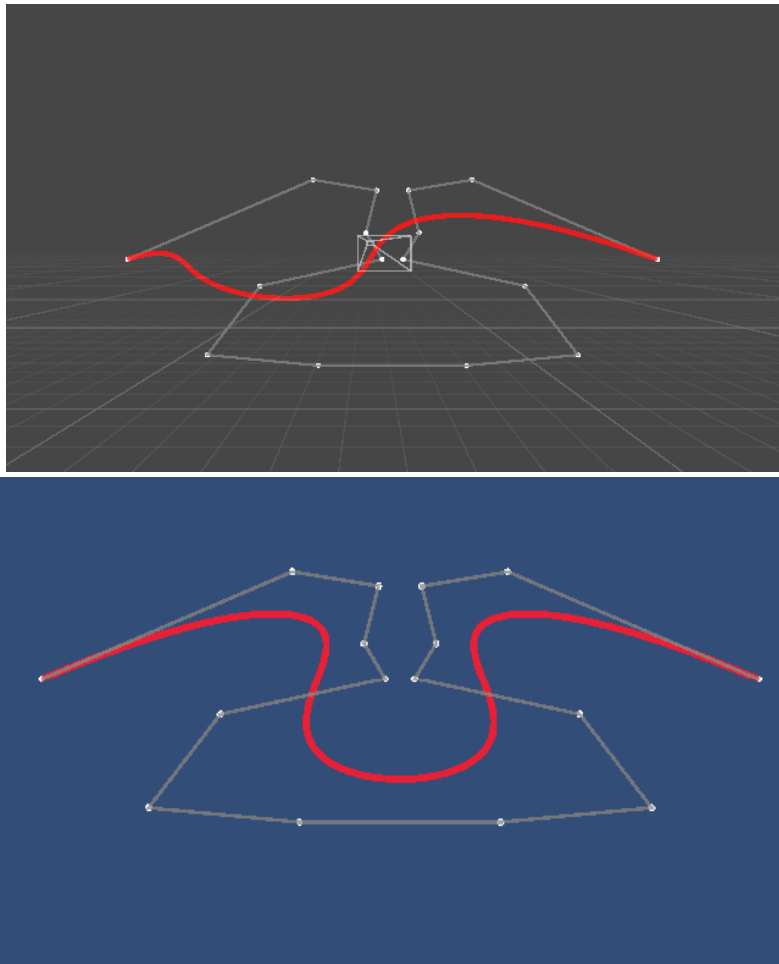


November 22, 2025

Today, I created a TempleBezierCurve script to implement a control point that form into a point that will later be developed on the Jigsaw Game.

However, I struck into a problem as the Bezier line wasn't the way displayed in the video I watched. Fortunately, I fixed it with the help of redoing the script again and as you can see

the Bezier line is more uniform than the other image above. Also, I was having a problem with the camera as the object wasn't in centre, but I just had to fix the position on the camera. Also, another problem I had was with the input manager was having multiple errors when ever I clicked on the screen. A way I solved it was to into Project Settings next Player then Other Settings and go to Active Input Handling and set it to Both. This fixed the problem easily and I continued with my work.



Week 4

November 25, 2025

Today, I created a sprite called Tile and TileGen as it'll implement FloodFill as it's a way to fix the tiles into place and will generate as a jigsaw tile.

Also, started the sprite called Board as it'll make a jigsaw board to make it more like a puzzle game.

November 26, 2025

Today, I went back on GitHub and tried to start typing up my Learning Log, but I couldn't as it didn't have a preview menu when I committed changes. I solved this problem by just doing the tutorials and learning log on word and just apply them into GitHub as PDF for people to see.

Then, I continued my GitHub project on the Tile script and just went over to see if there were any errors. Most errors were just misspellings as they are easily fixed with a quick review.

November 28, 2025

Today, I went onto my GitHub and create a TileMovement where you'll be able to pick up the tiles and move them around to make the image whole. Also, I added another script which is called TileSort which will sort out the tiles in random places.

Week 5

January 5th, 2026

Today, I started typing up my Tutorials in Word document then later to be apply the document into the GitHub file. Firstly, I started off with the first tutorial which is Implement Bezier Curve. After, a bit I finished the tutorial and I started a bit of the second tutorial which is to create a jigsaw tile.

January 6th, 2026

Today, I carried on making the tutorials and making changes to make it more understandable to the eye. Then, I updated my work log while adding descriptions in them.

January 7th, 2026

Today, I went back into my GitHub project and updated everything.

January 8th, 2026

Today, I implement all the tutorials on the GitHub and added a few notes to make it more refined. Also, I updated my Work Log and linked it into the GitHub.